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Bachelor Thesis Bioinformatics

Evaluation of LLMs and Knowledge Bases for Scientific Discovery

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25.02.2026

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Period: 27.10.2025 - 27.02.2026

Abstract

Large language models (LLMs) are increasingly used for biomedical science, yet their factual reliability remains difficult to quantify. This thesis introduces an evaluation framework that measures reliability as agreement with a biomedical knowledge graph (KG) baseline under repeated querying. We construct a controlled benchmark of 163 unique questions in the form of subject–object entity pairs by filtering 500 questions to retain only specific, simplified types of questions to ensure a controlled, reproducible evaluation by analysing harvested relations between the pairs.

For each entity pair, the LLM was prompted to produce relations in a structured *triple* format (subject–predicate–object), using only a restricted set of allowed relation types (Biolink predicates). Agreement with the KG reference was then measured using different analysis methods (precision, recall, and F_1). Results are reported in two ways: (i) *mean-over-runs*, which reflects typical single-query behavior averaged over repeated runs, and (ii) *union-over-runs*, which estimates how much can be recovered when the model is queried many times and all predicted triples are combined.

Across four prompt variants (v1–v4), exact agreement with the KG remained very low (macro- $F_1 < 0.03$), and many questions produced no matching relation at all. Even when each question was queried 100 times for the baseline prompt, union aggregation recovered only a small portion of the KG reference on average ($P_U \approx 0.094$, $R_U \approx 0.105$). Distributional analyses suggest that the gap is driven mainly by two failure modes: frequent abstention (empty outputs) and a strong concentration on a small number of relation types. Small prompt modifications measurably shifted abstention rates, output volume, and run-to-run stability, but did not substantially improve alignment with the KG baseline.

Overall, the proposed framework provides a reproducible, quantitative approach to assess LLM reliability for biomedical relation extraction, and highlights the importance of evaluating abstention behavior, semantic diversity, and run-to-run stability in addition to aggregate accuracy.

Acknowledgements

I would like to thank my supervisor, Professor Dr. Sven Nahnsen, for guidance, constructive feedback, and continuous support throughout the project, and for helping to shape the direction of this thesis. It is especially valuable for me, since this work is my first step into the world of Science.

Furthermore, I am grateful to my family members who made it possible for me to study in Germany and who were supportive, from my decision to change my major to bioinformatics, through writing this thesis. I want to thank them for their emotional support, which is very important during stressful and time-consuming periods.

In particular, I want to thank Dr. Khilola Rustamova, the chief of the department of Pediatric Oncology, Hematology, and Immunology in the Scientific and Practical Medical Center in Uzbekistan, who is also my mother. She consistently arouses my interest in the Biology part of my studies and lets me work with her on practical tasks related to her job in statistics and bioinformatics. She also helped me select some research questions and suggested methods of analysis.

I also thank the RENCI Translator/ROBOKOP community for providing the open biomedical knowledge graph infrastructure that made this evaluation possible.

Finally, I want to thank all my colleagues and friends for discussions and encouragement during the writing process.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
ARA	Autonomous Relay Agent
BERT	Bidirectional Encoder Representations from Transformers
Biolink	Biolink Model (biomedical data model used in Translator/ROBOKOP)
CLI	Command-Line Interface
CURIE	Compact URI (Uniform Resource Identifier)
F1	Harmonic mean of precision and recall
GTF	Generalized Text Format (generic term; if unused, remove)
GPT	Generative Pre-trained Transformer
JSON	JavaScript Object Notation
JSONL	JSON Lines (one JSON object per line)
KG	Knowledge Graph
LLM	Large Language Model
MACRO	Macro-averaging (average over questions)
MICRO	Micro-averaging (pooling over instances)
NGS	Next-Generation Sequencing
NLP	Natural Language Processing
P	Containment precision (precision-like agreement)
R	Containment recall (recall-like agreement)
ROBOKOP	Reasoning Over Biomedical Objects linked in Knowledge Oriented Pathways
SOTA	State of the Art
TSV	Tab-Separated Values
URI	Uniform Resource Identifier
NCATS	National Center for Advancing Translational Sciences

Chapter 1

Introduction

Large language models (LLMs) have rapidly become practical interfaces for accessing and synthesizing biomedical knowledge. Their ability to generate fluent text and structured output makes them attractive for tasks such as hypothesis generation, entity linking, and relation discovery. At the same time, biomedical applications are high-stakes: incorrect statements about gene–disease associations, drug mechanisms, or adverse effects can mislead downstream analyses and erode trust in computational pipelines. A central obstacle is that LLMs can produce statements that are plausible but not grounded in verifiable evidence—a phenomenon commonly discussed under the umbrella of hallucination and factual errors [5, 13]. This underscores the need for operational, reproducible reliability evaluations that go beyond anecdotal examples and quantify when, how often, and under what conditions an LLM produces verifiable biomedical relations.

Biomedical knowledge graphs (KGs) offer a complementary representation of biomedical knowledge: entities (e.g., diseases, genes, chemicals) are represented as nodes and relationships as typed edges. KGs can be used to support structured querying, evidence aggregation, and mechanistic reasoning. In translational science, the NCATS Biomedical Data Translator program has advanced a KG-centered ecosystem in which heterogeneous sources are semantically integrated and accessed through standardized interfaces [1, 3]. Within this ecosystem, ROBOKOP (Reasoning Over Biomedical Objects linked in Knowledge Oriented Pathways) provides a KG and a user-facing interface for exploring connections between biomedical entities and for question answering over integrated knowledge [9]. Interoperability across data sources and tools is supported by shared schemas such as the Biolink Model, which standardizes entity categories and predicate types across biomedical KGs [11].

Despite these advances, a practical methodological gap remains: *how reliable are LLM-generated biomedical relations when evaluated against a KG baseline under controlled constraints and repeated querying?* Two difficulties com-

plicate this question. First, “correctness” is not uniquely defined: KGs may be incomplete and reflect schema choices and snapshot coverage. Second, LLM outputs are stochastic and prompt-sensitive; therefore, a single response can be unrepresentative of typical behavior. Reliability assessment must therefore incorporate both *factual agreement with a structured baseline* and *robustness across repeated queries*, while accounting for sensitivity to prompt formulation.

1.1 Aim and research questions

This thesis proposes and evaluates an operational framework for assessing LLM reliability in biomedical relation discovery by comparing structured triples extracted from an LLM to a KG baseline (ROBOKOP) under repeated querying. The evaluation focuses on single-hop relations between biomedical entity pairs and uses a strict notion of agreement: a predicted triple is counted as correct only if subject, predicate, and object exactly match a deduplicated ROBOKOP baseline triple for the same entity pair.

The study addresses the following research questions:

- **KG agreement:** To what extent do LLM-extracted biomedical triples match a ROBOKOP baseline?
- **Robustness:** How stable are extracted relations across repeated queries of the same entity pair, and does aggregation over repeated runs increase baseline coverage?
- **Prompt sensitivity:** How do controlled prompt variants change abstention, output volume, predicate usage, semantic diversity, and agreement with the KG baseline?
- **Failure modes:** Which empirical mechanisms (e.g., abstention, predicate concentration) dominate the observed disagreement with the baseline?

1.2 Contributions and scope

The main contributions of this thesis are:

- A curated benchmark of 163 biomedical node pairs derived under constraints designed for automated evaluation (non-zero baseline signal, bounded baseline density, and single-hop framing).
- A reproducible evaluation pipeline that (i) constructs a deduplicated ROBOKOP baseline per node pair, (ii) queries an LLM under multiple

prompt variants with repeated runs, (iii) parses structured outputs into triples, and (iv) computes strict agreement metrics alongside diagnostic measures such as abstention rates, union-over-runs coverage, predicate frequency distributions, and stability statistics.

- An empirical analysis of reliability under strict KG-consistency, highlighting systematic failure modes and quantifying prompt-driven trade-offs without conflating output volume or stability with factual agreement.

The scope of the thesis is intentionally narrow in two respects. First, the benchmark and scoring focus on single-hop triples to enable unambiguous automated matching. Second, the ROBOKOP agreement is treated more as an operational baseline than as biomedical ground truth; the evaluation quantifies *consistency with a structured KG snapshot* of the latest available knowledge as of today, which may slightly change over time.

1.3 Thesis structure

The remainder of this thesis is organized as follows. Chapter 2 introduces biomedical knowledge graphs in translational science, the Translator/ROBOKOP ecosystem, and relevant concepts in LLM factuality and evaluation. Chapter 3 describes the benchmark construction, the prompting and repeated-query design, and the scoring methodology. The empirical results are presented in Chapter 4, including strict agreement metrics and diagnostic analyses of abstention, predicate distributions, union aggregation, and stability across prompt variants. Chapter 5 discusses the findings in context, outlines limitations, and proposes methodological improvements and future research directions.

Chapter 2

Background

This chapter provides the biological and methodological background needed to interpret the evaluation framework and results. It first summarizes how biomedical knowledge graphs support translational research and how interoperability standards shape KG content. It then reviews why LLM factuality is challenging to assess, especially for relation discovery, and introduces evaluation concepts that motivate the metrics used in this thesis.

2.1 Biomedical knowledge graphs in translational science

Biomedical knowledge graphs represent entities (e.g., genes, diseases, drugs, phenotypes) as nodes and encode relationships as typed, directed edges. In translational science, KGs are used to integrate heterogeneous sources, support mechanistic explanations, and facilitate structured queries for tasks such as drug repurposing and gene prioritization. The NCATS Biomedical Data Translator program was initiated to enable such integration at scale by organizing multiple teams and knowledge sources into a federated ecosystem for “translating” data into actionable biomedical insights [1, 3].

A core requirement for KG interoperability is a shared schema that defines entity categories and relation types. The Biolink Model addresses this need by providing an open, hierarchical schema for biomedical entities and predicates, enabling consistent interpretation of edges across knowledge sources and downstream tools [11]. Schema standardization is not merely a technical detail: predicate granularity and directionality determine how biomedical statements are represented and therefore what an “exact match” means in an automated evaluation.

Within the Translator ecosystem, ROBOKOP provides a KG-backed interface for exploring relationships and for question answering over integrated

biomedical knowledge [9]. ROBOKOP can return sets of triples connecting a given entity pair through one- or several-hop relations, which allows evaluation LLM-extracted triples under strict matching. Importantly, such a baseline is not necessarily synonymous with ground truth: KG content depends on source coverage, curation policies, schema choices, outdated information, and snapshot timing. For this reason, the present thesis interprets not only KG agreement as definitive of biomedical correctness, but also keeps in mind about imperfections of static Knowledge Graphs.

2.2 LLMs, relation discovery, and the problem of factuality

LLMs can produce structured relation hypotheses in natural language or in machine-readable formats (e.g., JSON triples). This capability is attractive for biomedical informatics because it potentially reduces manual effort in relation extraction and supports rapid hypothesis generation. However, LLM outputs can include unsupported claims or fabricated details, particularly in “closed-book” settings where the model is not required to cite evidence. Surveys on hallucination and factuality describe this as a persistent limitation of generative models and emphasize that fluent generation can mask factual errors [5, 13].

A common strategy to reduce factual errors is to incorporate retrieval or external evidence, for example through retrieval-augmented generation (RAG), where generation is conditioned on retrieved documents or structured context [6]. While retrieval-based approaches are promising, they also change the evaluation question: performance then depends on the retrieval system and the evidence provided. The present thesis therefore evaluates a closed-book extraction regime to isolate how reliably the model can recover KG-consistent relations without direct access to the KG during generation. This choice aligns with the goal of quantifying baseline agreement and robustness under repeated sampling rather than optimizing task performance with auxiliary tools.

2.3 Concepts for operational reliability evaluation

Reliability in this thesis is operationalized as *repeatable recovery of KG-consistent triples* under controlled constraints. Three methodological considerations motivate this definition.

Exact matching versus semantic plausibility. Automated evaluation requires a concrete match criterion. Exact triple-level matching (subject, predicate, object) is conservative: it penalizes near-misses such as semantically related predicates, inverse formulations, or alternative identifiers for the same biomedical concept. This strictness is useful when the goal is reproducible scoring and a high bar for KG consistency, but it can under-estimate broader semantic plausibility. Consequently, low agreement can arise from both genuine factual errors and schema-level mismatches. On the other hand, this does not apply to all relations. There are enough relations that cannot be mismatched. On a large scale, this would mean that some triples are unfairly penalized; there is another part that either has very high or at least some decent scores.

Stochastic generation and repeated querying. LLM outputs are not deterministic; repeated queries to the same prompt can yield different sets of predicted relations. Repeated-run designs therefore allow two complementary assessments: (i) *stability*, quantifying how consistent outputs are across runs, and (ii) *coverage under aggregation*, where union-over-runs measures whether repeated sampling recovers additional baseline relations beyond what is observed in a single run. Importantly, stability alone is not sufficient: a model can be stable because it frequently abstains (returns no edges) or because it repeats generic relations.

Output sparsity and concentration effects. Two empirical behaviors are particularly relevant for relation discovery. First, *abstention* (empty outputs) increases apparent stability but reduces utility and depresses mean agreement scores by construction. Second, *mode collapse* or output concentration can cause a model to overuse a small subset of generic predicates, reducing semantic coverage. Related phenomena have been discussed in the language-generation literature as degeneration and repetitive sampling behavior [4]. In a KG-consistency evaluation, predicate concentration can reduce strict agreement both by limiting expressivity (rare baseline predicates are never predicted) and by producing schema-level mismatches when generic association predicates substitute for more specific baseline relations.

These considerations motivate reporting agreement metrics (precision/recall/ F_1 under strict matching) together with diagnostic measures (non-empty rate, union coverage, predicate distributions, and stability). Taken together, they enable a more informative interpretation than any single score: they indicate whether low agreement is driven mainly by abstention, by concentrated predicate choices, or by broader inconsistencies with the baseline.

2.4 Summary

Biomedical knowledge graphs provide structured baselines for relation discovery and are supported by interoperability standards such as the Biolink Model within the Translator ecosystem [1, 11]. LLMs can generate candidate relations but remain susceptible to factuality errors and prompt-sensitive behavior [5, 13]. The evaluation framework in this thesis therefore treats reliability as a measurable, reproducible property defined by strict KG-consistent triple recovery under repeated querying, complemented by diagnostic analyses that separate agreement from abstention, concentration, and stability.

Chapter 3

Materials and Methods

This chapter describes the data sources, experimental design, model configuration, and analysis pipeline used to evaluate LLM reliability for biomedical relation discovery. The study is framed as a controlled agreement test between relations extracted by GPT and a knowledge-graph baseline extracted from RENCI ROBOKOP [2, 9]. Reliability is operationalized along three axes: (i) factual agreement with a KG baseline under explicit matching rules, (ii) robustness under repeated stochastic querying, and (iii) sensitivity to prompt formulation as a secondary intervention analysis [15].

3.1 Study design overview

The pipeline consists of eight stages:

1. **Benchmark construction:** generate candidate subject–object node pairs using GPT under a constrained harvesting prompt; retain only pairs that are single-hop queryable in ROBOKOP with bounded connectivity (1–10 edges); and deduplicate pairs to obtain a fixed benchmark.
2. **KG baseline extraction:** query ROBOKOP for each benchmark node pair and extract a baseline set of directed triples K of the form (subject_curie, predicate, object_curie).
3. **Baseline deduplication:** apply within-pair exact deduplication to remove repeated triples returned for the same pair.
4. **GPT relation extraction:** query GPT repeatedly for each node pair under prompt variants (v1–v4), enforcing a fixed JSON output schema and a restricted predicate vocabulary.

5. **Schema validation and filtering:** validate every GPT response against a strict output contract; reject malformed responses; validate and filter individual edges.
6. **Normalization and set construction:** transform baseline and GPT outputs into canonical directed triple sets per question and run.
7. **Scoring:** compute strict triple-level agreement metrics (containment precision/recall/F1), reported under macro and micro aggregation.
8. **Secondary analyses and reproducibility:** quantify output sparsity, predicate-frequency concentration (mode collapse), and run-to-run stability; store all intermediate artifacts (TSV/JSON/JSONL) to enable deterministic re-analysis without re-querying external services.

3.2 Data sources and benchmark construction

3.2.1 Entity representation

All biomedical entities are represented using CURIEs (e.g., MONDO:..., NCBIGene:..., CHEBI:...). Each benchmark item is a subject–object CURIE pair and is treated as one evaluation *question*. Human-readable labels are stored alongside CURIEs for interpretability.

3.2.2 From manual seeds to LLM-assisted harvesting

The project began with a small manually curated seed set (approximately 15 questions). This approach was insufficient because many candidate questions were not directly translatable into a consistent single-hop KG query format, were inherently multi-hop, yielded no ROBOKOP edges, or yielded too many edges (often > 10), undermining controlled evaluation.

To construct a larger and more uniform benchmark, candidate node pairs were generated using GPT with a constrained harvesting prompt that required: (i) a fixed tabular schema (question ID, labels, CURIEs, and notes) and (ii) that each pair be plausibly supported by ROBOKOP while targeting bounded connectivity. The harvesting prompt was executed five times, producing 500 candidate node pairs in total (Table 4.1).

3.2.3 ROBOKOP verification, connectivity filter, and node-pair deduplication

Candidates were filtered using ROBOKOP as a control mechanism:

- **Connectivity constraint:** retain only pairs for which ROBOKOP returned between 1 and 10 connecting edges.
- **Node-pair deduplication:** remove duplicate or effectively repeated node pairs introduced across harvesting runs.

After filtering, 163 unique node pairs remained and were held fixed for all subsequent experiments (Table 4.1).

3.3 Knowledge graph baseline extraction

3.3.1 ROBOKOP reference baseline

RENCI ROBOKOP integrates multiple curated biomedical sources and exposes relations as typed edges between entities [2]. For each benchmark node pair, ROBOKOP was queried to obtain a baseline set

$$K = \{(s, p, o)\},$$

where s and o are CURIEs and p is a predicate string aligned with the Biolink schema [11].

3.3.2 Within-pair exact deduplication

Because ROBOKOP aggregates evidence across multiple sources, raw results may contain identical triples within a single node pair. To obtain a consistent reference per pair, we applied *within-pair exact deduplication*: for each question, each distinct triple (s, p, o) was retained at most once, irrespective of multiplicity in the raw output. This preprocessing choice is consistent with standard KG integration practice, where redundant edges with identical endpoints and predicates are coalesced while provenance is handled separately [14]. The resulting baseline counts are reported in Table 4.3.

3.4 GPT relation extraction experiments

3.4.1 Model configuration

All experiments were conducted via API access using **GPT-5.2** under a fixed decoding and output configuration:

- **Temperature:** 0.7
- **Maximum output length:** 512 tokens

- **No retrieval augmentation:** no web search, no database lookup, no external context beyond the provided prompt
- **No tool use:** model-only generation (no function calling / tool invocation)

A nonzero temperature was used intentionally. First, a central study axis is robustness under repeated stochastic querying; near-deterministic decoding would suppress variability and weaken the ability to measure run-to-run stability. Second, many real-world interactive uses employ nonzero sampling to balance helpfulness and diversity; evaluating at $T = 0.7$ targets this regime and allows union-over-runs coverage estimates to be meaningful [10, 12]. In addition, repeated sampling remains relevant even when temperature is reduced, since practical systems can exhibit nondeterminism beyond nominal decoding settings [8].

3.4.2 Controlled inputs and prompt variants

For each benchmark question, GPT was queried under four prompt variants (v1–v4). Variants differ only in limited instruction framing and evidence requirements and are evaluated on the identical set of 163 node pairs, enabling direct comparison of sensitivity to prompt formulation [15]. Across variants, the following constraints are held constant:

- **Fixed input nodes:** each query specifies exactly two nodes by CURIE (the benchmark subject and object).
- **Restricted predicate vocabulary:** GPT is instructed to select predicates exclusively from an allowed list (Section 3.4.4).
- **Structured JSON output:** GPT must return a machine-readable JSON object with an `edges` list (Section 3.5).

Full prompt templates (harvesting and v1–v4 extraction prompts) are provided in the Appendix.

3.4.3 Repeated querying schedule

To quantify variability under stochastic decoding, each question was queried repeatedly. The baseline prompt (v1) was executed 100 times per question (16300 runs) to obtain high-resolution distributions. Prompt variants v2–v4 were executed 25 times per question (4075 runs per variant) as a prompt-sensitivity study; this repetition budget reflects a practical tradeoff between coverage and cost while still supporting stable aggregate estimates.

3.4.4 Predicate restriction and enforcement

To ensure direct comparability to ROBOKOP edges and to avoid unmatchable free-text relations, the predicate space was restricted to the **75 Biolink predicates** exposed by ROBOKOP for this task setting [11]. The allowed list was injected into the prompt as an explicit enumeration and treated as a hard constraint.

Enforcement occurred during parsing:

- Any predicted edge with a predicate not in the allowed set was discarded.
- After filtering, a run may become empty (**edges** list length 0), in which case it is treated as abstention ($|G| = 0$) for scoring.

This restriction (i) eliminates paraphrased predicates, (ii) reduces the hypothesis space to KG-compatible relations, and (iii) supports strict triple-level overlap without synonym mapping.

3.5 Output contract, schema validation, and run handling

3.5.1 JSON output contract

Each GPT response was required to be a single JSON object with the fields:

- `question_id`
- `subject_curie`, `object_curie`
- `edges`: a list of zero or more edge objects

Each edge object must contain `subject`, `predicate`, and `object` (CURIE strings and an allowed predicate string). If no relation is considered plausible under constraints, GPT must return `"edges": []`.

3.5.2 Schema validation and invalid runs

Programmatic schema validation was applied to every response because **LLMs** can violate strict structured-output constraints [7]. Validation checks included:

- JSON parseability (well-formed JSON)
- presence and type of required top-level fields

- `edges` must be a list
- each edge must contain required keys with string-typed values

Responses failing validation were rejected, logged with an error reason, and handled conservatively as abstentions by setting the parsed edge set to empty ($|G| = 0$) for downstream aggregation and scoring. This avoids inflating agreement by selectively excluding difficult runs and preserves a fixed denominator for repeated-run analyses.

3.6 Normalization and set construction

Before scoring, baseline and GPT triples were transformed into canonical triples and represented as sets:

$$(s, p, o).$$

Normalization includes trimming whitespace, enforcing consistent string types, and using the predicate string as provided (after allowed-list filtering). Duplicate predicted edges within a run are removed by set construction. No synonym expansion, ontology reasoning, or semantic similarity matching was applied; agreement is evaluated under strict exact matching on directed triples.

3.7 Agreement metrics

Agreement is quantified using containment-style precision and recall, computed on sets of directed triples. Let G denote the predicted triple set for one run and K the deduplicated ROBOKOP baseline triple set for the corresponding question.

Precision-like containment:

$$P(G, K) = \begin{cases} 0, & |G| = 0, \\ \frac{|G \cap K|}{|G|}, & \text{otherwise.} \end{cases} \quad (3.1)$$

Recall-like containment:

$$R(G, K) = \begin{cases} 0, & |K| = 0, \\ \frac{|G \cap K|}{|K|}, & \text{otherwise.} \end{cases} \quad (3.2)$$

F1 score:

$$F1(G, K) = \begin{cases} 0, & P(G, K) = 0 \wedge R(G, K) = 0, \\ \frac{2P(G, K)R(G, K)}{P(G, K) + R(G, K)}, & \text{otherwise.} \end{cases} \quad (3.3)$$

We additionally report the *zero-F1 rate*: the fraction of questions whose *mean* F1 over repeated runs equals 0 (i.e., no matching triple is produced across repetitions).

3.7.1 Macro vs micro reporting

Metrics are reported under two aggregation views:

- **Macro averaging:** compute per-question values (after run aggregation as defined below) and average across questions, giving each question equal weight.
- **Micro averaging:** pool counts (intersections and denominators) across questions and runs before computing the metric, weighting by triple volume.

3.8 Aggregation protocols over repeated runs

Because GPT outputs vary across repetitions, we report two complementary protocols.

3.8.1 Mean-over-runs (typical agreement)

For each question, compute $P(G_i, K)$, $R(G_i, K)$, and $F1(G_i, K)$ for each run $i = 1, \dots, n$, then average across runs to obtain per-question mean scores. These per-question values are then aggregated macro- and micro-wise across the benchmark. This protocol estimates typical single-run agreement for a given prompt variant.

3.8.2 Union-over-runs (coverage under repetition)

For each question, merge all predicted triples across runs into a single set:

$$G^{\cup} = \bigcup_{i=1}^n G_i.$$

Compute $P(G^{\cup}, K)$, $R(G^{\cup}, K)$, and $F1(G^{\cup}, K)$. This protocol estimates maximal coverage achievable by repeated querying and is especially informative under output sparsity.

3.9 Secondary analyses

3.9.1 Output sparsity and non-zero subset analysis

Output sparsity is quantified via the frequency of empty parsed outputs ($|G| = 0$), which can arise from explicit abstention (`edges: []`), from all edges being filtered out, or from invalid runs handled as abstention. We report:

- **Abstention rate:** fraction of runs with $|G| = 0$
- **Coverage (edge rate):** fraction of runs with $|G| > 0$
- **Mean predicted edges per run:** average $|G|$

To isolate correctness conditional on responding, we additionally perform a *non-zero subset* analysis (reported for v1): compute P , R , and $F1$ only over runs with $|G| > 0$, then aggregate macro and micro. We also compute a *match rate*: for each question, the fraction of runs with at least one exact match ($|G \cap K| > 0$), averaged over questions.

3.9.2 Predicate-frequency concentration (mode collapse)

To assess semantic diversity, we compute predicate-frequency distributions for (i) the deduplicated ROBOKOP baseline triples and (ii) GPT outputs aggregated across all questions and runs (per variant). For each distribution, we report:

- number of distinct predicates observed
- cumulative **top- k mass** for $k \in \{1, 2, 3, 5\}$ (fraction of all triples accounted for by the k most frequent predicates)
- identity and frequency of the top-5 predicates

High concentration (large top- k mass) is interpreted as evidence of reduced semantic diversity consistent with predicate mode collapse.

3.9.3 Run-to-run stability

Run-to-run stability is computed per question and summarized across questions. Reported quantities include:

- `edge_count_mean` and `edge_count_std` across runs

- `abstention_rate` across runs
- `predicate_set_stability`: similarity of the set of predicates across runs
- `triple_set_stability`: similarity of the full predicted triple set across runs

Predicate-set and triple-set stability are computed using average pairwise Jaccard similarity across runs:

$$J(S_i, S_j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|}.$$

For `predicate_set_stability`, S_i is the predicate set in run i ; for `triple_set_stability`, S_i is the triple set in run i . Stability is interpreted jointly with abstention rate and output volume.

3.10 Reproducibility and stored artifacts

All intermediate artifacts are stored in plain-text formats to enable deterministic re-analysis without re-querying external services:

- benchmark node pairs (TSV)
- ROBOKOP baseline triples (TSV), raw and within-pair deduplicated
- GPT outputs (JSONL/JSON), including raw responses and parsed edge lists
- per-run and per-question metrics (TSV/JSON summaries)
- generated figures (image files) included in Chapter 4

Apart from stochasticity inherent to model generation, all preprocessing, filtering, normalization, aggregation, and scoring steps are deterministic given the stored artifacts.

Chapter 4

Results

This chapter reports the empirical outcomes of the proposed evaluation framework for large language model (LLM) reliability in biomedical relation discovery. We first summarize how the final benchmark set of node pairs was obtained, then describe the ROBOKOP baseline extraction and its deduplication, and finally present quantitative agreement results between GPT outputs and the knowledge graph baseline across repeated runs and prompt variants. Where appropriate, we complement aggregate scores with distributional analyses (per-question variability), predicate-frequency analyses (semantic collapse), and stability measurements across repetitions.

4.1 Benchmark construction: from harvested node pairs to a controlled set of 163 questions

To obtain an evaluation set that is compatible with single-hop knowledge graph querying and robust statistical analysis, we generated candidate subject–object node pairs via GPT using a constrained harvesting prompt that required: (i) the output to follow a fixed tabular schema (question ID, labels, CURIEs, and notes), and (ii) each pair to be plausibly supported by ROBOKOP with a bounded number of connecting edges. The harvesting prompt was executed five times, producing 500 candidate node pairs in total. The candidates were then filtered using ROBOKOP as a control mechanism: only those pairs for which ROBOKOP returned between 1 and 10 connecting edges were retained. Finally, we removed duplicates introduced by GPT during the production of 500 node pairs (across generations). After these filtering steps, 163 unique node pairs remained and were used as the fixed benchmark throughout all subsequent experiments.

Table 4.1: Construction of the benchmark set. Candidate node pairs were harvested using GPT and then filtered using ROBOKOP connectivity constraints (1–10 edges) and deduplication.

Step	Count
Harvested candidate node pairs (5 runs)	500
After ROBOKOP connectivity filter (1–10 edges)	167
After node-pair deduplication	163

Table 4.2: Example benchmark question (one node pair).

Question ID	Subject (CURIE)	Subject name	Object (CURIE)	Object name
Q150	NCBIGene:1813	DRD2	CHEBI:3766	clozapine

4.1.1 Illustrative example question (node pair)

For interpretability, we include a concrete example of a benchmark question (one node pair) that is used consistently across the following qualitative examples.

4.2 ROBOKOP baseline extraction and deduplication

Each of the 163 node pairs was queried against RENCI ROBOKOP to obtain a knowledge-graph baseline consisting of triples of the form (`subject_curie`, `predicate`, `object_curie`). Because ROBOKOP integrates evidence from multiple underlying knowledge sources, its raw query results can contain repeated triples within the same node pair (for example, the same predicate between the same entities supported by multiple sources). To ensure a clean baseline for agreement evaluation, we applied within-pair exact deduplication, retaining each triple at most once per node pair.

This deduplication reduced the total number of baseline triples by approximately 20% across the benchmark.

4.2.1 Illustrative example of ROBOKOP output

Table 4.4 shows a representative excerpt of the baseline triples for the example node pair from Table 4.2. This is the reference set K used in the agreement evaluation.

Table 4.3: Effect of within-pair deduplication on ROBOKOP triples. Deduplication removes exact repeats of the same triple returned multiple times for the same node pair.

Quantity	Value
Total ROBOKOP triples (raw)	750
Total ROBOKOP triples (deduplicated)	605
Relative reduction	$\approx 20\%$
Mean baseline edges per question (deduplicated)	3.7117

Table 4.4: Example excerpt of deduplicated ROBOKOP baseline triples (K) for the example node pair, that originally had five copies of the `biolink:affects` predicate.

Subject (CURIE)	Predicate	Object (CURIE)
CHEBI:3766	<code>biolink:affects</code>	NCBIGene:1813
CHEBI:3766	<code>biolink:directly_physically_interacts_with</code>	NCBIGene:1813
CHEBI:3766	<code>biolink:related_to</code>	NCBIGene:1813
CHEBI:3766	<code>biolink:increases_response_to</code>	NCBIGene:1813
CHEBI:3766	<code>biolink:affects_response_to</code>	NCBIGene:1813

4.3 GPT extraction runs and stored outputs

For each benchmark node pair, GPT was queried repeatedly using prompts that enforced: (i) exactly two given nodes (by CURIE), (ii) a restricted predicate vocabulary (allowed predicates), and (iii) a machine-readable JSON structure suitable for automated parsing. If GPT determined that no relation was plausibly supported under the prompt constraints, it returned an empty `edges` list. All GPT outputs were stored in JSONL/JSON format for downstream aggregation and metric computation.

4.3.1 Illustrative example of GPT output

The following verbatim snippet illustrates the expected JSON structure for a single GPT response on the example node pair. This predicted triple set is denoted G . For our example node pair, GPT found no predicates.

```
{
  "question_id": "Q150",
  "subject_curie": "NCBIGene:1813",
  "object_curie": "CHEBI:3766",
  "edges": []
}
```

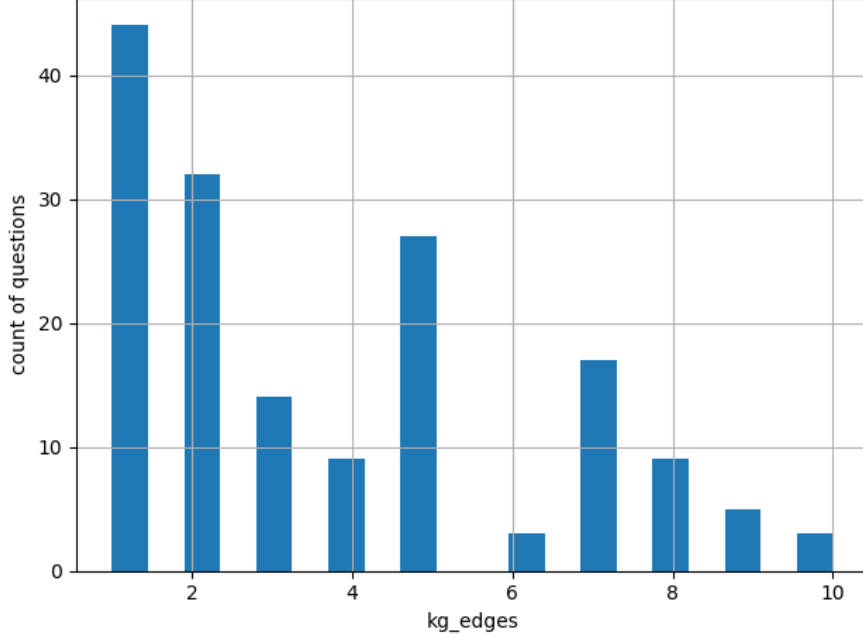


Figure 4.1: Distribution of deduplicated ROBOKOP baseline edges per question across the 163 node pairs.

4.4 Agreement metrics between GPT and ROBOKOP

We quantify agreement between GPT predictions G and the ROBOKOP baseline K using containment-style precision and recall, computed on sets of directed triples. Let $|G|$ be the number of predicted triples in a GPT output and $|K|$ the number of baseline triples for the corresponding node pair.

4.4.1 Precision-like and recall-like containment

We define precision-like containment as the fraction of predicted triples that are present in the baseline:

$$P(G, K) = \begin{cases} 0, & |G| = 0, \\ \frac{|G \cap K|}{|G|}, & \text{otherwise.} \end{cases} \quad (4.1)$$

We define recall-like containment as the fraction of baseline triples recovered by GPT:

$$R(G, K) = \begin{cases} 0, & |K| = 0, \\ \frac{|G \cap K|}{|K|}, & \text{otherwise.} \end{cases} \quad (4.2)$$

The per-output F1 score is the harmonic mean:

$$F1(G, K) = \begin{cases} 0, & P(G, K) = 0 \wedge R(G, K) = 0, \\ \frac{2P(G, K)R(G, K)}{P(G, K) + R(G, K)}, & \text{otherwise.} \end{cases} \quad (4.3)$$

Finally, we report the *zero-F1 rate*, defined as the fraction of questions whose *mean* F1 over repeated runs equals 0 (i.e., GPT never produced a matching triple for that node pair across the repetitions).

4.4.2 Two evaluation protocols: union-over-runs vs mean-over-runs

Because GPT responses vary across repeated queries, we report results under two complementary aggregation protocols:

- **Union-over-runs (per question):** for each question, we take the set-union of all predicted triples across runs, $G^\cup = \bigcup_{i=1}^n G_i$, and compute $P(G^\cup, K)$, $R(G^\cup, K)$, and $F1(G^\cup, K)$. This estimates *coverage* achievable by repeated querying.
- **Mean-over-runs:** we compute $P(G_i, K)$, $R(G_i, K)$, and $F1(G_i, K)$ per run and average them (macro-averaging over questions, and additionally micro-averaging over all predicted/baseline triples). This estimates the *typical* single-query agreement.

4.5 Overall agreement results and prompt-variant comparison

This section summarizes strict triple-level agreement between GPT outputs and the deduplicated ROBOKOP baseline across repeated runs and prompt variants. Results are reported under two aggregation protocols: (i) *union-over-runs*, which estimates how much baseline content can be recovered when repeatedly querying the model and pooling all predicted triples, and (ii) *mean-over-runs*, which reflects typical single-run performance by averaging per-run scores.

Across all prompts, agreement remained low. In addition, GPT frequently returned empty outputs (no edges), yielding high zero- F_1 rates. When non-empty outputs were produced, exact matches to baseline triples occurred for only a subset of questions, resulting in low recall-like containment under mean-over-runs aggregation.

Table 4.5: Baseline prompt (v1) summary under strict matching for 163 questions with 100 repeated GPT runs per question. Union-over-runs aggregates all predicted triples across runs before scoring (coverage upper bound), whereas mean-over-runs averages per-run scores (typical reliability).

Quantity	Value
Number of questions (n)	163
Zero- F_1 rate (mean-over-runs)	0.712
$\overline{P}_U$ (union precision-like containment)	0.094
$\overline{R}_U$ (union recall-like containment)	0.105
$\overline{F1}_U$ (union F_1)	0.085
$\overline{J}_U$ (union Jaccard)	0.054
$\overline{P}$ (mean precision-like containment)	0.108
$\overline{R}$ (mean recall-like containment)	0.032
$\overline{F1}$ (mean F_1)	0.045

4.5.1 Baseline prompt (v1): union-over-runs versus mean-over-runs performance

For the baseline prompt (v1), we executed 100 repeated GPT runs per question for $n = 163$ questions. Table 4.5 reports v1 performance under both aggregations. Union-over-runs merges all predicted triples across the 100 runs for each question into a single set before scoring against the ROBOKOP baseline; mean-over-runs computes scores per run and averages them.

Under union aggregation, GPT achieved $\overline{P}_U = 0.094$ and $\overline{R}_U = 0.105$ (Table 4.5), indicating limited recovery of baseline triples even after extensive repeated querying. The union recall distribution across questions is shown in Figure 4.2, which highlights substantial heterogeneity: some questions reach non-trivial union recall, whereas many remain near zero.

Under mean-over-runs aggregation, performance is lower, reflecting typical single-run behavior. In particular, mean recall-like containment dropped from $\overline{R}_U = 0.105$ under union aggregation to $\overline{R} = 0.032$ under mean-over-runs (Table 4.5), and mean F_1 similarly decreased from $\overline{F1}_U = 0.085$ to $\overline{F1} = 0.045$. The zero- F_1 rate was 0.712 (Table 4.5), for approximately 71% of questions no run produced any triple matching the ROBOKOP baseline.

4.5.2 Prompt variants (v1–v4): mean-over-runs comparison

To assess sensitivity to prompt formulation, we evaluated four prompt variants (v1–v4) that differ only in a small number of steering constraints and evidence requirements. For each node pair, agreement with the ROBOKOP baseline

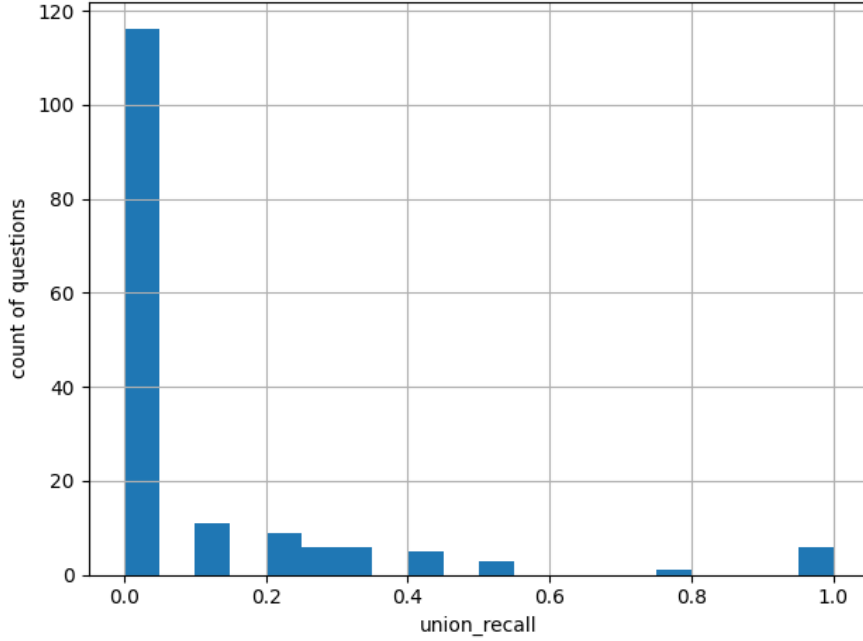


Figure 4.2: Distribution of union-over-runs recall across questions (v1).

Table 4.6: Agreement metrics for prompt variants under mean-over-runs aggregation.

Variant	Macro P	Macro R	Macro F_1	Zero- F_1 rate	Micro P	Micro R	Micro F_1
v1	0.07669	0.01897	0.02915	0.71166	0.13214	0.02408	0.04074
v2	0.09890	0.01654	0.02795	0.75460	0.13217	0.02664	0.04435
v3	0.03104	0.00562	0.00932	0.83436	0.14382	0.00846	0.01599
v4	0.08491	0.01544	0.02556	0.77301	0.20093	0.02288	0.04108

was computed per run and then aggregated as mean-over-runs. Variants v2–v4 were run 25 times per question (163 questions; 4075 total runs each), whereas v1 was run 100 times per question; consequently, union-based comparisons are only available for v1, and stability estimates across variants should be interpreted in light of the differing number of repetitions.

Table 4.6 reports macro-averaged containment precision, recall, and F_1 (averaged over questions), as well as micro-averaged metrics (pooled over all triples), and the zero- F_1 rate (fraction of questions whose mean F_1 over runs equals 0).

Across variants, macro- F_1 values remain low (all below 0.03). The highest macro- F_1 is observed for v1 (0.02915), followed by v2 (0.02795) and v4 (0.02556), while v3 yields the lowest macro- F_1 (0.00932). Macro precision

Table 4.7: Output volume by prompt variant (mean-over-runs).

Variant	Mean KG edges / question	Mean GPT edges / run
v1	3.7117	0.6764
v2	3.7117	0.7482
v3	3.7117	0.2184
v4	3.7117	0.4225

Table 4.8: Coverage and match rate by prompt variant (mean-over-runs).

Variant	GPT edge rate / question	Mean match rate / question
v1	0.5755	0.08717
v2	0.7482	0.09890
v3	0.2162	0.03141
v4	0.4226	0.08491

varies more strongly across prompts than macro recall: v2 yields the highest macro precision (0.09890), while v3 yields the lowest (0.03104). Zero- F_1 rates are high for all variants and range from 0.71166 (v1) to 0.83436 (v3).

Micro-averaged metrics show different rankings in places: v4 yields the highest micro precision (0.20093), while v2 yields the highest micro F_1 (0.04435).

In addition to agreement, Tables 4.7 and 4.8 summarize output volume and coverage. Table 4.7 reports the mean number of baseline edges per question and the mean number of GPT-predicted edges per run. Table 4.8 reports coverage (fraction of runs with a non-empty output) and a match-rate statistic (fraction of runs that contain at least one triple matching the baseline, averaged over questions).

Tables 4.7 and 4.8 show that output volume and coverage differ substantially by prompt. Coverage is highest for v2 (0.7482), followed by v1 (0.5755) and v4 (0.4226), while v3 has the lowest coverage (0.2162). The mean match rate follows a similar ordering: v2 is highest (0.09890), v1 and v4 are close (0.08717 and 0.08491), and v3 is lowest (0.03141).

4.6 Why are the scores low? Output sparsity and predicate mode collapse

Two empirical patterns account for most of the observed gap between GPT predictions and the ROBOKOP baseline in this evaluation: (i) frequent empty outputs (output sparsity) and (ii) concentration of predicted relations into a

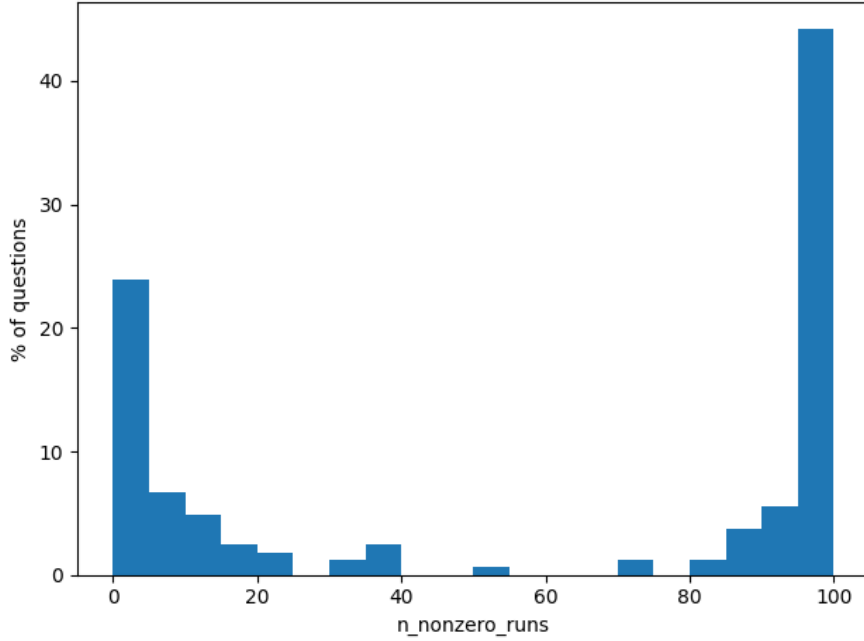


Figure 4.3: Distribution of the number of non-empty GPT outputs per question in v1 (out of 100 runs per question).

small subset of predicates (predicate mode collapse). This section summarizes these patterns and quantifies their effect for the v1 setting.

4.6.1 Output sparsity: frequency of empty outputs

Across repeated queries, GPT often returned an empty `edges` list. Under our metric definitions, empty outputs contribute $P(G, K) = 0$ by construction and typically yield $F1(G, K) = 0$, increasing the fraction of questions with mean F1 equal to zero.

To quantify how often GPT produces any relation at all, we report the number of non-empty runs per question. In v1, the experiment comprises 163 questions with 100 repeated runs each ($N_{\text{runs}} = 16300$). Of these, $N_{\text{nonzero}} = 9381$ runs contained at least one predicted edge, corresponding to an overall non-empty rate of $9381/16300 \approx 0.5755$. Averaged across questions, the mean number of non-empty runs was 57.55 per question.

Figure 4.3 shows that output sparsity is not uniform across questions: some questions receive non-empty outputs in only a small fraction of runs, whereas others receive non-empty outputs in most runs.

Table 4.9: v1 non-zero subset summary. Metrics are computed only on runs with non-empty GPT outputs ($|G| > 0$). Macro scores average per-question values; micro scores pool triples across all non-empty runs.

Quantity	Value
Questions total	163
Non-empty runs total ($ G > 0$)	9381
Mean non-empty runs per question	57.5521
Macro precision (non-zero)	0.10815
Macro recall (non-zero)	0.03189
Macro F1 (non-zero)	0.04505
Micro precision (non-zero)	0.13214
Micro recall (non-zero)	0.03739
Micro F1 (non-zero)	0.05829

4.6.2 Conditional agreement given non-empty outputs (v1 non-zero subset)

To separate the effect of abstention from the correctness of outputs when GPT does respond, we perform a *non-zero subset* analysis for v1. Here, we exclude runs with $|G| = 0$ and compute containment precision, recall, and F1 only over runs where GPT produced at least one edge. Formally, for each question, metrics are computed across its subset of runs satisfying $|G| > 0$ and then aggregated across questions (macro) and across triples (micro).

Table 4.9 summarizes the resulting agreement metrics. Relative to the unconditional mean-over-runs evaluation, the conditional (non-zero) scores are higher, reflecting that empty runs are excluded; however, the conditional agreement remains low overall.

The per-question distributions of mean precision, recall, and F1 under the non-zero subset are shown in Figure 4.4. All three distributions are strongly right-skewed with substantial mass near zero, indicating that many questions remain difficult even when conditioning on non-empty outputs, while a smaller subset of questions achieves higher agreement values.

In addition, we report a *match rate among non-empty runs*: for each question, the fraction of non-empty runs that contain at least one triple matching the ROBOKOP baseline ($|G \cap K| > 0$), averaged over questions. The distribution of this statistic is shown in Figure 4.5, which again indicates that questions vary substantially in how frequently any match is observed once GPT produces an output.

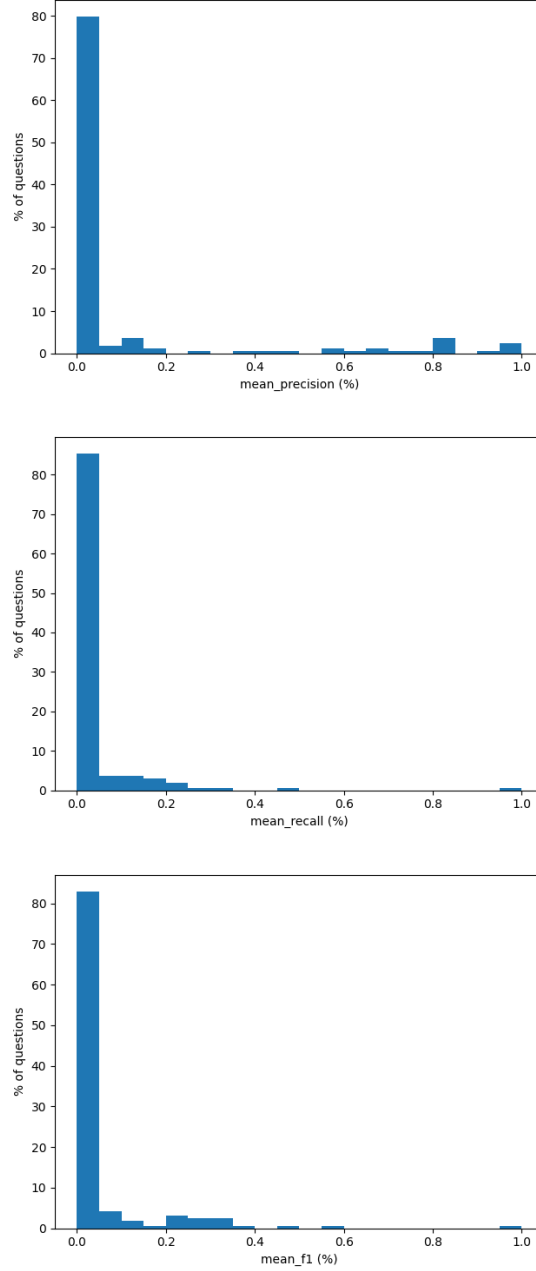


Figure 4.4: Per-question distributions of mean containment precision, mean containment recall, and mean F_1 in the v1 non-zero subset (computed only over runs with $|G| > 0$).

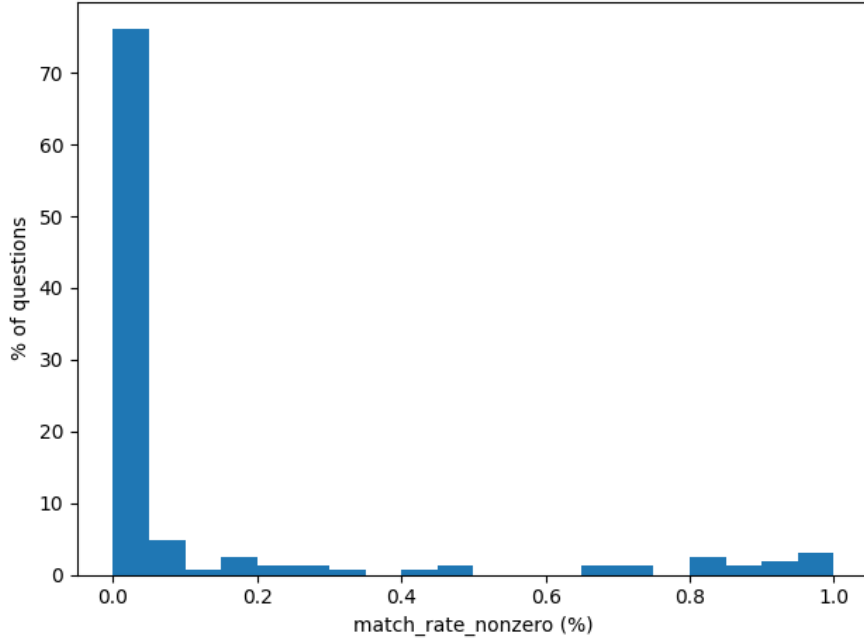


Figure 4.5: Distribution of the match rate among non-empty runs in v1. For each question, this is the fraction of runs with $|G| > 0$ that satisfy $|G \cap K| > 0$, shown as a percentage of questions.

4.6.3 Predicate mode collapse: reduced semantic diversity in GPT outputs

To characterize the semantic diversity of extracted relations, we computed global predicate-frequency distributions for (i) the deduplicated ROBOKOP baseline triples and (ii) GPT outputs aggregated over all questions and runs, separately for each prompt variant (v1–v4). For each source, we report the number of distinct predicates observed and the cumulative mass of the top- k predicates (top-1, top-2, top-3, top-5), which summarizes how concentrated the distribution is.

Table 4.10 shows that the ROBOKOP baseline is comparatively dispersed across many predicates (top-5 mass 64.6%), whereas GPT outputs are substantially more concentrated. In v1, the two most frequent predicates account for 76.0% of all predicted triples, and the top-5 account for 93.4%. Similar concentration is observed for v2–v4, with v3 and v4 exhibiting the highest top-1 mass (61.9% and 58.0%, respectively).

To make the identity of dominant predicates explicit, Table 4.11–4.12 lists the top-5 predicates (with fractions) for ROBOKOP and for each GPT variant. Notably, the ROBOKOP baseline also includes ontology-structural and intervention-related relations among its most frequent predicates (e.g.,

Table 4.10: Predicate distribution concentration (top- k cumulative mass in percent).

Source	# pred.	Top-1	Top-2	Top-3	Top-5
ROBOKOP (dedup)	22	18.0	32.9	46.1	64.6
GPT v1	25	40.9	76.0	83.8	93.4
GPT v2	16	38.3	58.3	76.6	93.2
GPT v3	8	61.9	78.7	91.7	98.3
GPT v4	6	58.0	88.7	96.0	99.9

`subclass_of` and `treats_or_applied_or_studied_to_treat`), while GPT distributions are dominated by disease–gene association and interaction predicates.

4.7 Stability across repeated GPT queries

One of the central goals of this thesis is to treat run-to-run variability under repeated querying as an operational notion of LLM reliability. We therefore quantify stability across repetitions using two complementary perspectives: (i) *output-volume stability*, capturing how consistently GPT produces a non-empty set of triples, and (ii) *semantic stability*, capturing how similar the predicted relation sets are across repeated runs.

For each prompt variant (v1–v4), stability statistics were computed per question over repeated runs and then aggregated across the 163 questions. The reported per-question quantities are: (i) `edge_count_mean` (mean number of predicted edges per run), (ii) `edge_count_std` (standard deviation of predicted edge counts across runs), (iii) `abstention_rate` (fraction of runs with an empty `edges` list), (iv) `predicate_set_stability` (similarity of the set of predicates across runs), and (v) `triple_set_stability` (similarity of the full predicted triple set across runs). Table 4.13 reports the mean and median (across questions) for each quantity.

Across variants, output volume and abstention differ substantially. Variant v2 exhibits the highest mean `edge_count_mean` and the lowest mean `abstention_rate`, indicating that it produces non-empty outputs most consistently. In contrast, v3 exhibits the lowest output volume and the highest abstention rate, meaning that many runs return no triples at all. Variant v4 lies between these regimes, while v1 shows intermediate abstention and output volume.

Semantic stability metrics (`predicate_set_stability` and `triple_set_stability`) are higher for v3 and v4 than for v2 (Table 4.13). At the same time, v3 and v4 exhibit higher abstention rates and lower output

Table 4.11: Top-5 predicates by frequency (ROBOKOP, GPT v1, GPT v2). Fractions are in percent.

Rank	ROBOKOP (dedup)	GPT v1	GPT v2
1	<code>interacts_with</code> (18.0)	<code>gene_associated_</code> <code>with_condition</code> (40.9)	<code>gene_associated_</code> <code>with_condition</code> (38.3)
2	<code>affects</code> (14.9)	<code>interacts_with</code> (35.1)	<code>directly_</code> <code>physically_</code> <code>interacts_with</code> (20.0)
3	<code>subclass_of</code> (13.2)	<code>physically_</code> <code>interacts_with</code> (7.8)	<code>interacts_with</code> (18.3)
4	<code>genetically_</code> <code>associated_with</code> (9.8)	<code>genetically_</code> <code>associated_with</code> (6.3)	<code>physically_</code> <code>interacts_with</code> (14.7)
5	<code>treats_or_</code> <code>applied_or_</code> <code>studied_to_treat</code> (8.8)	<code>directly_</code> <code>physically_</code> <code>interacts_with</code> (3.2)	<code>produces</code> (1.9)

volumes than v2. Therefore, stability summaries are reported together with `abstention_rate` and `edge_count_mean`.

4.8 Prompt wording as a measurable intervention

The prompt variants differ only by a small number of lines, yet they produce measurable differences in output sparsity, predicate usage, agreement, and stability. Differences in agreement are summarized in Table 4.6, output volume and coverage in Table 4.7–4.8, predicate-frequency concentration in Table 4.10, and run-to-run stability in Table 4.13.

In terms of output sparsity, variants separate into regimes with distinct abstention rates. Under the stability summaries (Table 4.13), v2 shows the lowest mean abstention rate (0.252), whereas v3 shows the highest (0.784), with v1 and v4 intermediate (0.424 and 0.577). This pattern is consistent with the output coverage summaries reported earlier (Table 4.7–4.8).

Agreement metrics remain low across all variants (Table 4.6), with macro-F1 values below 0.03. The relative ordering differs depending on macro vs

Table 4.12: Top-5 predicates by frequency (GPT v3, GPT v4). Fractions are in percent.

Rank	GPT v3	GPT v4
1	gene_associated_with_ condition (61.9)	gene_associated_with_ condition (58.0)
2	physically_interacts_with (16.7)	interacts_with (30.8)
3	interacts_with (13.0)	physically_interacts_with (7.3)
4	directly_physically_interacts_with (5.1)	directly_physically_interacts_with (2.5)
5	disease_has_basis_in (1.6)	disease_has_basis_in (1.4)

Table 4.13: Stability summary across questions (mean (median) over 163 questions). Columns: μ_e = edge_count_mean, σ_e = edge_count_std, a = abstention_rate, S_p = predicate_set_stability, S_t = triple_set_stability.

Variant	μ_e	σ_e	a	S_p	S_t
v1	0.676 (1.010)	0.328 (0.369)	0.424 (0.110)	0.768 (0.805)	0.768 (0.805)
v2	0.748 (1.000)	0.099 (0.000)	0.252 (0.000)	0.690 (0.703)	0.661 (0.653)
v3	0.218 (0.040)	0.183 (0.000)	0.784 (0.960)	0.831 (1.000)	0.815 (0.920)
v4	0.423 (0.280)	0.144 (0.000)	0.577 (0.720)	0.832 (1.000)	0.764 (0.847)

micro aggregation: v1 achieves the highest macro-F1 (0.02915), v2 achieves the highest micro-F1 (0.04435), and v3 yields the lowest values under both views. Macro scores are computed per question and averaged across questions, whereas micro scores pool all triples across all questions and runs before computing the metric.

Stability metrics show additional variation across variants (Table 4.13). Mean triple-set stability is highest for v3 (0.815) and lowest for v2 (0.661), while mean predicate-set stability is highest for v4 (0.832) and lowest for v2 (0.690). Together, these results quantify that small prompt modifications are associated with measurable shifts in (i) abstention frequency, (ii) the concentration of predicate usage (Section 4.6.3), and (iii) the stability profile of outputs across repeated runs.

4.9 Summary of key findings

In summary: (i) controlled harvesting and ROBOKOP filtering yielded a fixed benchmark of 163 unique node pairs with 1–10 deduplicated baseline triples per pair; (ii) within-pair deduplication reduced the raw ROBOKOP baseline from 750 to 605 triples ($\approx 19\text{--}20\%$), resulting in an average of 3.71 baseline edges per question; (iii) under strict triple-level matching with mean-over-runs aggregation, GPT–ROBOKOP agreement remained low across all prompt variants (macro- $F_1 < 0.03$) with high zero- F_1 rates; (iv) two dominant empirical patterns accompanied low agreement: frequent abstention (empty outputs) and predicate-frequency concentration consistent with mode collapse; and (v) small prompt modifications measurably shifted abstention and stability profiles, indicating that prompt wording acts as an intervention, but did not close the alignment gap to the KG baseline under strict matching.

Chapter 5

Discussion

This thesis developed and applied an operational evaluation framework to quantify the reliability of large language models (LLMs) for biomedical relation discovery. Reliability was defined in a deliberately stringent, but also the best-suited informatics-oriented way: repeated extraction of structured subject–predicate–object triples that match a knowledge-graph (KG) baseline (ROBOKOP) for a controlled benchmark of 163 biomedical node pairs. In the following, the results from Chapter 4 are interpreted with two goals: (i) to explain the dominant empirical patterns in terms of concrete failure modes and methodological choices, and (ii) to place these findings into biomedical context, clarifying how to assess biomedical correctness of the GPT5.2.

5.1 Main findings in relation to the thesis aim

Across all prompt variants (v1–v4), strict triple-level agreement with the deduplicated ROBOKOP baseline remained low, with macro- F_1 below 0.03 and high zero- F_1 rates (Table 4.6). Under the baseline prompt (v1), repeated querying combined with union aggregation improved coverage only marginally ($\overline{P}_U = 0.094$, $\overline{R}_U = 0.105$; Section 4.5.1). From the standpoint of the operational definition used here—*repeatable recovery of KG-consistent triples*—these results indicate that the evaluated model GPT5.2 is not a reliable extractor of relations between a large fraction of biomedical questions.

Meanwhile, prompt wording acted as a measurable intervention on several dimensions relevant to reliability: output sparsity (abstention), output volume, predicate diversity, and run-to-run stability (Tables 4.7–4.13). These behavioral shifts did not translate into substantial gains in strict baseline agreement. The overall picture is therefore not only low accuracy, but a set of systematic, interacting failure modes that persist across prompt variants: frequent abstention, strong semantic concentration into a small predicate set, and limited exploration of baseline-consistent alternatives even under repeated sampling.

5.2 Interpreting low strict agreement: interacting informatics and biomedical factors

Several mechanisms plausibly contribute to low exact agreement. Some are inherent to the benchmark definition (strict matching against a specific KG snapshot), whereas others reflect model behavior under constrained output requirements.

Benchmark complexity is bounded but heterogeneous. The benchmark was designed to avoid both zero-signal node pairs and extremely dense neighborhoods by selecting pairs with 1–10 deduplicated baseline edges (Table 4.1). Nevertheless, baseline sizes somehow vary (Figure 4.1). Under strict recall, this may have some influence. A larger $|K|$ requires multiple correct extractions to achieve non-trivial recall, while for small $|K|$ a single correct edge can produce moderate recall. The observed recall distributions (e.g., Figure 4.2) remain concentrated near zero across much of the benchmark, suggesting that low agreement cannot be attributed solely to baseline-size effects.

Abstention is a dominant driver of mean performance and is question-dependent. A major empirical contributor to low mean-over-runs scores is that the model frequently returns an empty `edges` list, which yields $P = R = F_1 = 0$ by construction (Section 4.4). In v1, the overall non-empty rate is moderate on average, but per-question behavior is highly heterogeneous (Figure 4.3). This heterogeneity indicates that abstention is not just a random noise; it probably depends on entity types, the perceived plausibility of expressing relations using the restricted predicate vocabulary, and the model’s internal uncertainty under no external references allowed settings.

From a biomedical perspective, abstention is ambiguous. It may reflect conservative behavior when evidence is weak or when relations are context-dependent (e.g., conditional on tissue, exposure, or study design). However, for biomedical discovery workflows, frequent abstention is also a failure mode because it produces no actionable hypotheses. Reliability, in this sense, requires evaluating abstention jointly with correctness: a system that is “stable” because it often outputs nothing is not necessarily reliable for generating biomedical candidates.

Conditional correctness remains low even when outputs are non-empty. Conditioning on $|G| > 0$ increases agreement (as expected) but does not eliminate the core issue. For v1, macro precision and recall in the non-zero subset increase yet remain low, with many questions still producing baseline-matching triples only rarely (Table 4.9; Figure 4.4). This suggests

that disagreement is not solely explained by abstention. Instead, even when the model produces relations, it often selects subjects/objects/predicates that do not match the baseline under exact criteria.

In biomedical terms, two interpretations must be separated. First, the model may generate relations that are biomedical-sounding but not supported by ROBOKOP (hallucination relative to the baseline). Second, in some cases, the model may generate relations that are plausible or even true in the literature but absent from the baseline due to KG incompleteness, schema choices, or snapshot effects. The present study can decisively quantify the former only in the narrow sense of non-agreement with ROBOKOP; it cannot, by itself, adjudicate biomedical truth without external evidence.

Predicate mode collapse limits semantic coverage and increases schema-level mismatches. Predicate-frequency analyses show that model outputs are substantially more concentrated than the ROBOKOP baseline (Table 4.10). While the deduplicated baseline distributes predicate mass more broadly, the model concentrates heavily on a small number of generic predicates (top-5 mass $\geq 93\%$ across variants), and later variants use only a handful of predicates despite a larger allowed vocabulary.

This matters for strict matching in at least two ways. First, mode collapse prevents the model from expressing baseline relations that require less frequent predicates. Second, even when the underlying biomedical relation is directionally or conceptually similar, selecting a generic association predicate (e.g., `gene_associated_with_condition`) instead of the specific baseline predicate produces an exact mismatch. Under the strict evaluation used here, this is counted as disagreement even if the prediction is arguably “close” at a higher semantic level. This distinction is important for biomedical interpretation: disagreement may reflect schema coarsening rather than complete semantic failure.

KG-specific factors: canonicalization, identifiers, and baseline scope. ROBOKOP operates within the Translator ecosystem and aggregates signals from multiple underlying sources, with its own canonicalization rules and Biolink predicate semantics. Exact matching therefore requires agreement not only on biomedical meaning but also on identifier choice (CURIE normalization), relation directionality, and predicate granularity. Any residual mismatch in canonicalization between the baseline and LLM outputs—for example, producing an alternative CURIE for the same concept or choosing an inverse predicate representation—is penalized as an error by design. The strictness is appropriate for the operational reliability definition, but it also implies that the reported F_1 values are conservative lower bounds on broader “semantic” agreement.

5.3 Repeated querying and union aggregation: limited gains and what they imply

Union-over-runs was used to test whether repeated sampling can recover baseline edges by treating the model as a stochastic generator of candidate triples. Under v1, union aggregation increased coverage compared to mean-over-runs recall, but the gain remained limited ($\bar{R}_U = 0.105$; Section 4.5.1), and the distribution of union recall stayed strongly concentrated near zero (Figure 4.2). This indicates that repeated querying mainly re-samples within a restricted region of the model’s output space rather than exploring a diverse set of baseline-consistent alternatives.

In informatics terms, this is an important negative result: naive repetition does not behave like an effective search strategy for retrieving KG-consistent triples under the constraints of this task. In biomedical terms, it suggests that repeated prompting alone is unlikely to provide robust coverage of known KG relations for many entity pairs, limiting its usefulness as a stand-alone hypothesis generator when consistency with a curated KG is required.

5.4 Prompt interventions: trade-offs without closing the agreement gap

The prompt variants were designed as controlled interventions that emphasize distinct behaviors (e.g., encouraging output vs encouraging caution), rather than a full optimization over prompt space. The results nonetheless reveal interpretable trade-offs that are directly relevant to reliability:

Coverage vs. conservatism (abstention). Variants that reduce abstention increase the number of produced edges (Table 4.7) but do not necessarily increase strict agreement (Table 4.6). This suggests that additional output often consists of non-matching relations, which may correspond to baseline-inconsistent hypotheses or schema-level mismatches.

Stability vs. informativeness. Higher stability (Table 4.13) can coincide with lower output volume and higher abstention. Therefore, stability must not be interpreted as reliability in isolation: a stable system that consistently outputs empty lists, or repeats a small set of generic predicates, may be predictable but not useful for biomedical discovery.

Macro vs. micro behavior and uneven performance across questions. Micro metrics can appear comparatively better when correct matches are con-

centrated in a subset of questions or runs that contribute many pooled triples, whereas macro metrics penalize broad failure across the benchmark. The persistent low macro- F_1 across variants therefore indicates that agreement failures are widespread and not limited to a small subset of difficult cases.

Overall, the prompt interventions confirm high behavioral sensitivity to wording (a robustness concern in itself), but they do not resolve the core limitation: strict KG-consistent triple recovery remains rare for many node pairs in this no access to resources settings.

5.5 Limitations and implications for generalisation

Several limitations constrain what can be concluded beyond the operational setting studied here:

ROBOKOP is a structured, high-quality baseline, but not absolute biomedical ground truth. Disagreement with ROBOKOP implies non-agreement with this KG snapshot and its schema choices; it mostly but not always implies biomedical incorrectness. Conversely, agreement strongly implies KG-consistency rather than independent validation. This limitation is intrinsic to KG-based evaluation and motivates complementary validation strategies (e.g., literature evidence or cross-KG confirmation) in follow-up work.

Exact matching is conservative. The framework requires exact matches in subject, predicate, and object identifiers. Semantically close predicates, valid inverses, and alternative canonical identifiers are scored as mismatches. This is appropriate for a strict informatics definition of reliability, but it likely underestimates semantic plausibility and can conflate schema mismatch with factual mismatch.

Single-hop framing restricts biomedical expressivity. The benchmark focuses on single-hop edges. Many biomedical mechanisms are inherently multi-hop (e.g., drug $\rightarrow$ target $\rightarrow$ pathway $\rightarrow$ phenotype) and context-dependent. Thus, low performance indicates limitations in strict single-hop KG-consistent extraction without retrieval.

Stability can be inflated by abstention and repetition. Stability metrics can increase when outputs are often empty or when the model repeats a small set of generic relations. Reporting stability jointly with abstention and

output volume (as done here) is therefore necessary to avoid over-interpreting stability as correctness.

Despite these limitations, the evaluation remains informative for the thesis aim: it provides a reproducible, automated, and strict operationalization of reliability as KG-consistent extraction under repeated querying, and it supplies diagnostic signals (sparsity, union coverage, predicate concentration, stability) that explain *why* strict agreement is low.

5.6 Outlook: methodological and model-level improvements

The results motivate several concrete improvements for future work, aimed at separating schema effects from biomedical effects while preserving reproducibility:

- **Add graded matching layers.** In addition to strict exact matching, report relaxed evaluations (e.g., predicate equivalence classes, inverse predicate handling when appropriate, ontology-aware identifier equivalence) to disentangle schema-level mismatches from substantive disagreement.
- **Model abstention explicitly.** Treat abstention as a first-class outcome and evaluate selective prediction trade-offs (coverage vs correctness), rather than absorbing abstention into mean scores alone. This aligns better with biomedical use cases where returning “no claim” may be preferable to unsupported relations.
- **Condition stability on non-empty outputs.** Report stability both unconditionally and conditional on $|G| > 0$, to avoid attributing stability to correct relation extraction when it is driven by empty outputs.
- **Introduce evidence constraints via retrieval.** If the target is KG consistency, retrieval-augmented prompting (e.g., providing candidate predicates, candidate nodes, or supporting KG fragments) may reduce hallucinated predicates and improve alignment. This would move the setting from closed-book extraction toward evidence-grounded extraction, which is often the more relevant regime in biomedical informatics.
- **Stratify errors by biomedical relation type.** Analyze performance by entity-type combinations (disease–gene, drug–gene, disease–drug, etc.) and by baseline predicate categories to identify whether failures are uniform or concentrated in specific biomedical relation classes.

- **Evaluate against multiple KGs and/or literature evidence.** Repeating the same framework across alternative biomedical KGs, and validating a subset of predictions against external evidence sources, would help separate model limitations from baseline incompleteness and KG-specific schema effects.

5.7 Conclusion

Using a strict and reproducible KG-consistency definition of reliability, this thesis found low exact agreement between LLM-extracted biomedical triples and a deduplicated ROBOKOP baseline across 163 controlled node pairs. Performance was dominated by frequent abstention, strong predicate mode collapse, and limited gains from repeated querying, indicating that naive repetition does not constitute an effective strategy for recovering baseline-consistent triples in this closed-book setting. Prompt variants produced measurable and interpretable trade-offs in output volume and stability, but they did not substantially close the agreement gap. These findings suggest that future reliability evaluations should jointly model abstention, correctness, and stability, and incorporate graded matching and evidence constraints to distinguish strict schema-level mismatch from broader biomedical plausibility while maintaining automated, scalable scoring to get an even better understanding of the extent to which this and other LLMs are suitable for biomedical scientific research.

Appendix A

Further Tables and Figures

Appendix A

Appendix

This appendix contains supplementary material that supports reproducibility and provides additional detail that would otherwise interrupt the main narrative (Chapters 2–5). In particular, it documents the exact prompt texts used for all LLM runs, and lists supporting resources used by the evaluation pipeline (e.g., allowed predicate vocabulary, per-question metric tables, and extended diagnostic outputs).

A.1 Full prompt texts (v1–v4)

The following prompts were used verbatim to generate GPT outputs in the experiments reported in Chapter 4. Each prompt was executed under the same general extraction setup: the model received exactly two entities identified by their CURIEs (subject and object), and was required to return a JSON object with an `edges` list containing zero or more triples using only the allowed predicate vocabulary.

A.1.1 Prompt v1 (baseline)

Extract biomedical relations between exactly two given nodes. Return only relations that plausibly hold. Use only the allowed predicates and only the provided CURIEs. Return the result as JSON.

A.1.2 Prompt v2 (single most plausible relation)

Extract biomedical relations between exactly two given nodes. Return the single most plausible relation based on general biomedical knowledge, but only if at least one relation is plausibly supported. If none are plausibly supported, return an empty edges list. Use only the allowed predicates and only the provided CURIEs.

A.1.3 Prompt v3 (curated KB constraint)

Extract biomedical relations between exactly two given nodes. Return only relations that are commonly asserted in curated biomedical knowledge bases. If no such relation is commonly asserted, return an empty edges list. Use only the allowed predicates and only the provided CURIEs.

A.1.4 Prompt v4 (single most commonly asserted curated relation)

Extract biomedical relations between exactly two given nodes. Return the single most commonly asserted relation in curated biomedical knowledge bases, if any such assertion exists. If none exist, return an empty edges list. Use only the allowed predicates and only the provided CURIEs.

A.2 Allowed predicate list used for constrained extraction

This section lists the allowed predicate set used to constrain GPT outputs throughout the experiments. The vocabulary was derived from the Biolink predicate space used in the Translator/ROBOKOP ecosystem and is provided to enable exact replication of the extraction constraints.

A.3 Reproducibility notes

All code and analysis scripts used to generate the results are available in the project repository:

<https://github.com/azuziz/llm-vs-kg-evaluation>

Parts of the implementation were drafted and iteratively refined with assistance from GPT-5.2 (ChatGPT). All code was reviewed and executed by the author to generate the reported results.

The repository contains scripts for querying GPT and ROBOKOP, pre-processing and deduplication utilities, and evaluation and plotting code. To reproduce the main results, the required inputs are the final benchmark node-pair list, the deduplicated ROBOKOP baseline triples, and the saved GPT JSONL outputs for each prompt variant. The computational environment (Python version and package dependencies) is documented in the repository.

Table A.1: Allowed Biolink predicates used in all prompt variants (75 total). Long predicate names are listed at the end in full-width rows.

biolink:active_in	biolink:actively_involved_in
biolink:acts_upstream_of	biolink:affects
biolink:affects_response_to	biolink:ameliorates_condition
biolink:applied_to_treat	biolink:associated_with
biolink:capable_of	biolink:catalyzes
biolink:causes	biolink:chemically_similar_to
biolink:coexists_with	biolink:coexpressed_with
biolink:colocalizes_with	biolink:composed_primarily_of
biolink:contraindicated_in	biolink:contributes_to
biolink:correlated_with	biolink:decreases_response_to
biolink:derives_from	biolink:develops_from
biolink:diagnoses	biolink:disease_has_basis_in
biolink:disrupts	biolink:enables
biolink:expressed_in	biolink:gene_product_of
biolink:genetically_interacts_with	biolink:has_active_ingredient
biolink:has_adverse_event	biolink:has_input
biolink:has_metabolite	biolink:has_output
biolink:has_part	biolink:has_participant
biolink:has_phenotype	biolink:homologous_to
biolink:in_complex_with	biolink:in_taxon
biolink:increases_response_to	biolink:interacts_with
biolink:is_frameshift_variant_of	biolink:is_missense_variant_of
biolink:is_nearby_variant_of	biolink:is_non_coding_variant_of
biolink:is_nonsense_variant_of	biolink:is_splice_site_variant_of
biolink:is_synonymous_variant_of	biolink:located_in
biolink:manifestation_of	biolink:negatively_correlated_with
biolink:occurs_in	biolink:overlaps
biolink:physically_interacts_with	biolink:positively_correlated_with
biolink:precedes	biolink:predisposes_to_condition
biolink:preventative_for_condition	biolink:produces
biolink:regulates	biolink:related_to
biolink:similar_to	biolink:subclass_of
biolink:target_for	biolink:temporally_related_to
biolink:treats	biolink:catalyzes
biolink:acts_upstream_of_or_within_positive_effect	
biolink:acts_upstream_of_or_within_negative_effect	
biolink:directly_physically_interacts_with	
biolink:acts_upstream_of_positive_effect	
biolink:acts_upstream_of_negative_effect	
biolink:gene_associated_with_condition	
biolink:genetically_associated_with	
biolink:treats_or_applied_or_studied_to_treat	

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Selbständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Bachelorarbeit selbständig und nur mit den angegebenen Hilfsmitteln angefertigt habe und dass alle Stellen, die dem Wortlaut oder dem Sinne nach anderen Werken entnommen sind, durch Angaben von Quellen als Entlehnung kenntlich gemacht worden sind. Diese Bachelorarbeit wurde in gleicher oder ähnlicher Form in keinem anderen Studiengang als Prüfungsleistung vorgelegt.

Ort, Datum

Tübingen, den 27.02.2026

Unterschrift


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